Annant Pathak

ic24btech11002@iith.ac.in | annantpathak@gmail.com | +91 6261162593 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.31
Sundar Lal shrivastava H.S school	Class XII (MP)	2023	84.20%
St. joseph convent sr. sec. school	Class X (CBSE)	2021	88.20%

SCHOLASTIC ACHIEVEMENTS

MotiveVerse AI Hackathon 2nd Place GitHub: Engineered LLM-powered query system using Claude (AWS Bedrock) for video metadata extraction; implemented domain-specific prompt rules and error-correction logic for traffic classification

Solved 3**00+ DSA** problems on platforms including LeetCode and GeeksforGeeks.

EXPERIENCE

UG Research Assistant (ML)

Chemistry Department, IIT Hyderabad

- Built datasets for chemical structures such as MOFs and COFs for computational modeling tasks.
- Applied data preprocessing, regression modeling, and visualization using Python (Pandas, NumPy, Seaborn, scikit-learn)

Marketing Lead

Tayog

- Drove user acquisition from 100-200 to 1000+ signups over 3 months across 15+ colleges; achieved Top 5 in startup pitch competition validating product-market fit
- Recruited and managed 5-15 cross-functional ambassadors spanning IITs/NITs/IISERs; designed distributed outreach model enabling simultaneous multi-campus activation with coordinated strategy alignment

PROJECTS

FeedRank: C++ Social Feed System Github

- Designed and deployed production-grade social feed ranking system using C++, SQLite, and Crow REST framework; implemented Max Heap-based ranking algorithm achieving O(n log k) time complexity vs. O(n log n) sorting, improving system throughput by 10x
- Built normalized database schema with prepared statements to prevent SQL injection; strategically denormalized like/comment counts for read optimization; containerized with Docker multi-stage builds for efficient cloud deployment

Pathfinding Algorithm Visualizer (C++, SFML) Github

- Developed interactive pathfinding visualizer in C++ using SFML; implemented 4 algorithms (Dijkstra, BFS, DFS, A*) with customizable grid sizes, adjustable animation speed, and real-time algorithm step visualization for algorithm comparison
- Engineered maze management system supporting interactive obstacle placement, file-based configuration persistence (save/load maze templates), and dynamic path rendering

LLM From Scratch (PyTorch) Github

Built a GPT-2 style decoder-only transformer from scratch in PyTorch covering the complete ML pipeline:

- Implemented tokenization, causal multi-head self-attention, and feedforward networks
- Trained on custom text corpus with full pretraining loop (loss computation, optimization, gradient updates)

- Fine-tuned model on instruction-following tasks and implemented sampling-based text generation
- Architected reusable PyTorch modules (MultiHeadAttention, TransformerBlock, GPTModel) with proper abstractions

E-Commerce Platform Github

- Built a full-stack e-commerce platform with user authentication, cart management, order history, and profile pages, using Flask, SQLAlchemy, and SQLite with CSRF-protected forms via Flask WTF.
- Integrated the Razorpay payment gateway for secure transaction processing, handling client-side checkout and server-side payment verification via webhook/API routes.
- Designed a custom dark-themed responsive UI with Bootstrap 4, implementing an Amazon-style dual purchase flow.

POSITIONS OF RESPONSIBILITY

Head Positions: Web Dev Head | Alumni Cell || Head | Blockspace - Blockchain Club

Core Positions: Core | Blockspace - Blockchain club || Coordinator | OCS - Career Cell || Coordinator | Cultural Council

SKILLS

Languages: C++, Python, C, SQL

ML/AI: PyTorch, Transformer Architectures, scikit-learn, Pandas, NumPy

Tools & Backend: Docker, Git, Linux, SQLAlchemy, SQLite, Flask, REST API Design, Crow Framework, SFML

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Algorithm Complexity Analysis, Query Optimization

RELEVANT COURSES

Mathematics: Calculus, Differential Equations, Probability, Elementary Linear Algebra

Computer Science: Introduction to Artificial Intelligence, Introduction to Programming